

AGENTIC STAFFING PLATFORM

1. Problem Statement

Staffing and recruiting agencies generate new business by finding companies that are actively hiring and reaching out before competitors do. Today, this process is largely manual: a business development rep scans job boards, company news, LinkedIn, and press releases by hand, then makes a subjective judgment call about whether a company is worth pursuing as a client.

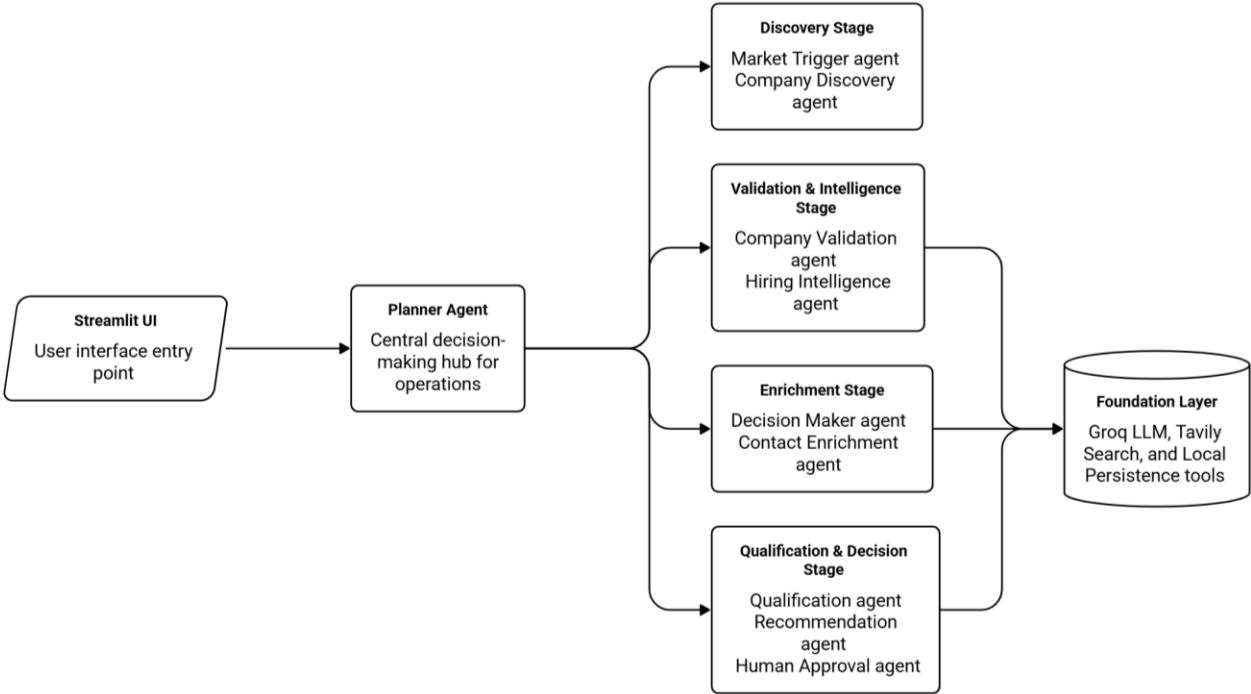
This platform automates company-level lead discovery and qualification for staffing sales teams. It is explicitly not a job search tool — job postings are an input signal, never the output. The output of every run is a ranked list of companies, each with a qualification rationale, hiring-intelligence context, and (where available) a relevant decision-maker contact — ready for a BD rep to act on.

2. System Architecture

The platform is built as a multi-agent system orchestrated with LangGraph, rather than a single large prompt or a fixed linear pipeline. A Planner Agent decides, at each step, which specialized agent should run next based on the current state of the research — using LangGraph's `Command(goto=...)` routing rather than a hardcoded sequence.

Agent	Role
Planner Agent	Decides which agent runs next based on current state; the only agent with explicit routing authority.
Market Trigger	Detects market-level hiring signals (e.g. funding events, expansion news) that suggest a wave of companies worth investigating.
Company Discovery	Generates the initial candidate list of companies from market triggers and search.
Company Validation	Confirms a discovered company is real, active, and matches basic targeting criteria (e.g. employee size limits) before further spend on research.
Hiring Intelligence	Determines what roles a company is hiring for, how urgently, and what that implies about staffing need. Primary driver of lead quality.
Company Enrichment	Gathers firmographic detail — industry, size, location, funding stage — to support qualification.
Decision Maker	Identifies a plausible decision-maker (e.g. Head of Talent, HR Director) at the company.

Contact Enrichment	Attempts to find contact details for the identified decision-maker.
Qualification	Scores the company against qualification criteria and produces a written rationale. Second primary depth area.
Recommendation	Synthesizes findings into a recommended outreach angle for the sales rep.
Human Approval	Holds the lead for manual review before it is marked Approved or Rejected.



3. Tools

Tool / Library	Role in the Platform
Python	Core implementation language for the entire platform.
LangGraph	Multi-agent orchestration; implements dynamic routing via the Planner Agent's Command pattern.
Groq (llama-3.3-70b-versatile)	LLM provider for all agent reasoning.
Tavily	Sole search backend for company discovery, validation, and enrichment.

Streamlit	Multipage web UI for configuring runs, monitoring progress, and reviewing/approving companies.
Local file-based persistence	Stores approved/rejected company decisions between runs.

4. Setup

Prerequisites

- Python 3.10 or later installed locally.
- A Groq API key (console.groq.com).
- A Tavily API key (tavily.com).

Installation Steps

1. Unzip or clone the project folder.
2. Create and activate a virtual environment :

```
python -m venv venv && source venv/bin/activate
```
3. Install dependencies:

```
pip install -r requirements.txt
```
4. Fill in your keys (set GROQ_API_KEY and TAVILY_API_KEY):

```
.env
```

Running

Go to u folder and run

```
cd ui  
streamlit run app.py
```